GyroMorpho v2

User Manual

A web-based pipeline for morphometric analysis
and automated taxonomic description of Gyrodactylidae

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https://github.com/wboeger/AI_morpho

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1. Overview

GyroMorpho v2 is a web-based pipeline for morphometric analysis of sclerotized structures of Gyrodactylidae (Monogenoidea). It uses landmark-based geometric morphometrics with Generalized Procrustes Analysis (GPA) to:

- Extract 2D pseudolandmarks from specimen images using ImageJ macros
- Align specimens via GPA to remove size, position, and orientation differences
- Compute discrete character states from continuous shape measurements
- Generate species descriptions and taxonomic diagnoses automatically
- Export phylogenetic matrices in Nexus, TNT, CSV, and JSON formats

The pipeline handles five structure types: marginal hooks, anchors, superficial bars, deep bars, and male copulatory organs (MCO). It ships with 36 pre-defined characters (C01-C12 for hooks, A01-A09 for anchors, B01-B06 for bars, D01-D03 for deep bars, M01-M06 for MCO).

2. Installation

Requirements

- Python 3.10 or later
- pip (Python package manager)
- ImageJ or Fiji (for landmark extraction macros)
- A modern web browser (Chrome, Firefox, Safari, Edge)

Setup

1. Clone the repository:

```
git clone https://github.com/wboeger/AI_morpho.git
cd AI_morpho
```

2. Create a virtual environment (recommended):

```
python3 -m venv venv
source venv/bin/activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Initialize the data directory:

```
mkdir -p data/uploads
```

5. Run the application:

```
python run.py
```

Open http://127.0.0.1:5000 in your browser. Register an account (the first user becomes admin), then create a project.

3. Pipeline Workflow

The project dashboard shows a Pipeline bar at the top with the complete workflow. Follow these steps in order:

1. **ImageJ Macros** — Download and run the ImageJ landmarking macros to extract 100 equidistant pseudolandmarks from specimen images.
2. **Import Landmarks** — Import the CSV files produced by the macros into the web application. Also import part boundary JSON files and specimen images.
3. **Assign Boundaries** — Use the interactive boundary editor to define which landmarks belong to which anatomical parts (Point, Shaft, Toe, etc.).
4. **Define Characters** — Review and customize character definitions, geometric operations, state thresholds, and dependencies in the Character Workshop.
5. **Compute & Review** — Batch-compute all geometric characters using Procrustes alignment. Review the character matrix, use the gallery to assign or correct states.
6. **Descriptions & Diagnoses** — Generate automated species descriptions and comparative taxonomic diagnoses.
7. **Export** — Export the final character matrix in Nexus, TNT, CSV, or JSON format for phylogenetic analysis.

4. Step 1: Extract Landmarks with ImageJ

Two ImageJ macros are included in the macros/ directory. They are also downloadable from the Import page with pre-configured directories for your project.

Hook Macro (macrogyrolandmark_v5.5.ijm)

Landmarks placed interactively:

- L1 = Point tip (tip of the hook)
- L2 = Toe tip (tip of the toe)
- L3 = Junction Point-Shaft (inner face)

Anchor Macro (macrogyrolandmark_v5_anchors.ijm)

Landmarks placed interactively:

- L1 = Point (tip of the anchor)
- L2 = External tip of the superficial root
- L3 = Distal-most base of the deep root

How to Run

1. Download the macro from the Import page (or find it in the macros/ folder).
2. Open ImageJ/Fiji.
3. Go to Plugins > Macros > Run... and select the .ijm file.
4. Configure the session: set input directory (folder with images), output directory (where CSVs will be saved), and processing options.

Macro Workflow (11 stages)

1. Session setup: configure directories, enhancement settings, landmark count (100), wand tolerance.
2. Image review: accept, reject (with logged reason), or skip each image.
3. Crop & upscale: draw bounding rectangle around the structure; 3x upscale for precision.
4. Enhancement: optional Gaussian blur, CLAHE contrast normalization, Unsharp Mask.
5. Orientation: verify the structure points to the right; optional horizontal flip.
6. B&W conversion: optional threshold-based binary mask for cleaner contour extraction.
7. Wand tool: click on the structure outline to extract the contour; adjustable tolerance.
8. Contour smoothing: 3-point moving average (configurable number of passes).
9. Landmark placement: click to place L1, L2, L3 sequentially with visual confirmation.
10. Equidistant resampling: 100 points starting from L1, evenly spaced by arc length.
11. Verification & editing: review color-coded overlay (cyan=L1, yellow=L2, magenta=L3, green=semilandmarks); optionally drag points to correct positions.

Output: one CSV file per specimen (X,Y columns, 100 rows) plus QC and rejection log files. Full backward navigation is available at every stage.

5. Step 2: Import Data into GyroMorpho

From the project page, click "Import from Folders". The import page is organized into four sections.

Import Landmarks (CSV files)

1. Click "+ Add Folder".
2. Enter the full path to the folder containing CSV files (output from the ImageJ macros).
3. Select the structure type (Marginal Hook, Anchor, etc.).
4. Click "Scan" to preview files and parsed species names.
5. Add more folders as needed (one per structure type).
6. Click "Import All".

Species names are automatically parsed from filenames. Supported formats: Gyrodactylus_salaris.csv (underscores), AB063294Gyrodactylusanguillae.csv (accession prefix), JF836137.1|Gyrocerviceanseris.csv (pipe-separated).

Import Part Boundaries (JSON files)

If you have pre-existing boundary definitions, import them as JSON files that map specimen names to part indices (1-based). Example:

```
{ "Gyrodactylus_salaris": {
    "Point": [1, 2, 3, 4, 5],
    "Shaft": [6, 7, 8, 9, 10]
  }
}
```

After boundary import, character states are automatically computed.

Import Images

Enter the path to a folder containing specimen images (PNG, JPG, GIF). Images are matched to existing specimens by species name from the filename. Use "Scan" to preview matches before importing.

6. Step 3: Assign Part Boundaries

On the project page, each structure has a "boundaries" link (visible when landmarks exist). The boundary editor assigns landmark points to anatomical parts.

Structure Parts

Structure	Parts
Hook	Point, Shaft, Toe, Shelf, Base, Heel
Anchor	Point, Shaft, SuperficialRoot, DeepRoot
Superficial Bar	BarProper, Shield, ShieldDistalEnd, AnterolateralProcesses
Deep Bar	(single unit)
MCO	Bulb, PrincipalSpine, Spinelets

Editor Modes

- Click mode: click individual landmarks to assign them to the active part.
- Range mode: click two landmarks to assign the entire range between them.
- Lasso mode: draw a freehand selection around landmarks.

Keyboard Shortcuts

- 1-6: select parts; C/R/L: switch modes; Ctrl+Z: undo
- "Copy from similar": automatically copies boundaries from the most morphologically similar specimen that already has confirmed boundaries.

Click "Confirm" to save boundaries and trigger automatic character computation.

7. Step 4: Define and Edit Characters

Click "Character Workshop" from the project page. Here you manage all character definitions.

Workshop Features

- View all characters grouped by structure type.
- Toggle characters active/inactive (inactive ones are excluded from the matrix and exports).
- Create new characters with custom geometric operations or manual coding.
- View value distributions to check threshold placement.

Edit Character Page

Clicking a character opens a two-panel editor:

Left panel (form):

- Name, description, parts involved, geometric operation, formula.
- States with drag-and-drop reordering, up/down arrows, and delete buttons.
- Each state has: code, name, description, threshold min, threshold max.
- Manual (gallery-coded) characters: the threshold columns are hidden, leaving a clean code / name / description editor. States are fully editable here - change, add, or delete.
- Dependencies: conditions under which the character is inapplicable.

Right panel (reference):

- Grid of all specimen thumbnails for this structure type, showing state badges and raw values.
- Structure type dropdown to view other structures for context.
- Shapes/Images toggle button.
- Click any thumbnail to open a lightbox with enlarged view.

Measurement explanation:

A collapsible box at the top of the edit page explains exactly how the system computes the character: which geometric operation is used, what it measures on the landmark coordinates, how Procrustes alignment is applied before measurement, and how raw values are mapped to discrete states via thresholds.

8. Step 5: Character Matrix and Gallery

Character Matrix

Click "Character Matrix" from the project page:

- Rows = specimens (species), columns = characters.
- Cells are color-coded by confidence: green (high), yellow (medium), red (low), gray (inapplicable).
- Click any cell to see details (raw value, confidence, computation type) and override the state.
- Filter by structure type (Hook, Anchor, Bar, MCO, All).
- Filter by DNA-only specimens or unconfirmed-only.

Gallery View

Click a character code in the matrix header to open the gallery:

- Specimens sorted by raw value (geometric characters) or state (manual characters).
- Landmark-derived shape outlines colored by anatomical parts, with a color legend.
- Structure type switcher: view shapes/images for other structure types as context.
- Shapes/Images toggle: switch between landmark outlines and uploaded photographs.
- Lightbox: click any thumbnail to zoom in, with state assignment buttons.
- Inline state buttons: assign states directly from the gallery grid.
- "Edit states" button (top of the gallery): opens a small editor to change, add, or delete the character's states without leaving the coding workspace. Codes must be unique; existing geometric thresholds are preserved by code. The page reloads so the new state buttons and badges appear immediately.

Tip: use the gallery to visually compare specimens side by side and assign states for manual characters or correct automatic assignments.

9. Step 6: Species Descriptions and Diagnoses

Species Descriptions

Click "Descriptions" from the project page. The system auto-generates morphological descriptions for each specimen based on its character states, formatted in standard taxonomic prose. Click "Regenerate" to update after changing character values.

Taxonomic Diagnoses

Click "Diagnoses" from the project page. Create taxonomic groups (genus, subfamily) by selecting which species belong to each group. The system generates comparative diagnoses highlighting distinguishing features. Diagnoses can be edited manually.

10. Step 7: Export

Click "Export" from the project page. Available formats:

Format	Description	Use case
CSV	Simple matrix (species x characters)	Spreadsheet analysis
CSV Detail	Matrix with raw values and confidence	Detailed analysis
Nexus	Standard phylogenetic format	MrBayes, PAUP*, Mesquite
TNT	TNT format	TNT parsimony analysis
JSON	Complete project data	Backup, re-import
Descript.	Formatted species descriptions	Publications
Diagnoses	Formatted group diagnoses	Publications

All matrix exports support optional filters: structure type, DNA-only specimens.

11. How Character States Are Computed

Generalized Procrustes Analysis (GPA)

Before computing character values, all specimens of the same structure type are aligned using GPA:

1. Center: each specimen's landmarks are translated so the centroid is at the origin.
2. Scale: landmarks are scaled to unit centroid size (removes size differences).
3. Rotate: specimens are iteratively rotated to minimize the sum of squared distances to the mean shape.

This ensures that character measurements are scale-independent and orientation-independent, capturing pure shape variation.

Geometric Operations

Operation	Description	Example
ratio_arc_length	Ratio of arc lengths of two parts	C01: Point/Shaft
sinuosity	Arc length / chord length	C03: Point waviness
mean_curvature	Mean Menger curvature along a part	C05: Shaft curvature
max_curvature	Maximum local curvature	Sharpest bend
junction_angle	Angle at junction between parts	C02: Point-Shaft angle
direction_angle	Angle between direction vectors	C06: Shaft-Base angle
relative_position	Normalized vertical displacement	C04: Point vs Toe
presence_thresh.	Part arc as fraction of total	C10: Heel presence
sinuosity_w_dir	Signed sinuosity (in/outward)	C04: Point direction
angle_betw_parts	Angle at fork between parts	A09: Shaft-root angle

State Mapping

Each raw numeric value is mapped to a discrete state using threshold ranges defined in the character definition. For example, if a junction angle is < 80 degrees it may be state "1" (recurved), 80-140 degrees state "2" (approximately 90 degrees), and > 140 degrees state "0" (evenly curved). A confidence score is computed based on distance from the nearest threshold boundary: farther from boundary = higher confidence.

12. Collaboration: Sharing Projects and Comments

GyroMorpho projects can be shared with other registered users, and specimens can carry a running thread of comments for team communication.

Sharing a Project

On the project page, the "Members & Sharing" panel lists everyone with access. To share:

1. In the share box, type the other user's username or email. An autocomplete list suggests existing users as you type.
2. Choose a role: Annotator (code data), Reviewer (review/confirm), or Admin (full control, including managing members).
3. Click "Share". The project immediately appears on that user's dashboard.

Any member can share a project. The project owner and Admin members can also remove members with the "Remove" button. The owner is marked with an "owner" badge and cannot be removed. Each member is shown with their username, email, and role.

Roles are labels for coordinating teamwork; the owner and Admins manage who has access. Removing a member revokes their access to the project immediately.

Specimen Comments

On the project (Specimens) page, every specimen row has a comment button showing a speech bubble and a count. Use comments to leave notes for collaborators - questions about an image, identification doubts, coding decisions, and so on.

- Click the comment button to open the thread for that specimen.
- Type in the box and click "Post" (or press Ctrl/Cmd + Enter). Each comment records its author and timestamp.
- Delete your own comments with the small x; Admins can delete any comment.
- The count on the button updates live as comments are added or removed.

13. Data Storage and Backup

Where Data Lives

- data/db.sqlite - SQLite database containing all structured data: specimens, structures, landmark coordinates, part boundaries, character definitions, character values, DNA sequences, descriptions, diagnoses, and activity logs. This is the single source of truth.
- data/uploads/ - Specimen images (PNG, JPG, GIF) copied during image import.

Important Notes

- The original CSV landmark files from ImageJ are NOT stored as files in the application. Their coordinates are parsed at import time and saved into the database. The original CSVs remain wherever you pointed the importer at.
- Export routes read directly from the database and generate output files on the fly.
- To back up a project, copy the entire data/ directory.

Backup tip: periodically copy the data/ folder to a safe location. The db.sqlite file contains everything except the images.

14. Molecular Phylogenetic Analysis

The Phylogeny module estimates a molecular tree from DNA sequences and imports it into the project so the morphological matrix and the phylogeny share one set of taxa. It runs as a background job: sequences are retrieved from GenBank (or uploaded), aligned, trimmed, and analysed by maximum likelihood, with two expert-review checkpoints along the way. Open it from a project via the Phylogeny page.

Markers and Analysis Modes

- Single marker: 18S rRNA, ITS (ITS1-5.8S-ITS2), 28S, COI, COII, or a custom Entrez query.
- Concatenated 18S + ITS: each marker fetched, aligned, and trimmed independently, then concatenated by species (taxa missing one marker are gap-filled).
- 18S-guided concatenation: 18S is retrieved first; ITS is then fetched for the union of the 18S species and the Specimens-page selection, so an ITS-only specimen is still kept.
- Multi-fragment: any combination of the five markers, analysed as a partitioned matrix with a per-fragment substitution model.

Starting a Job

On the Phylogeny page, set:

- Target taxon (e.g. Gyrodactylidae) and the per-marker Entrez gene query.
- NCBI email (required by Entrez). An NCBI API key is configured server-side to raise the request rate.
- Minimum sequence length and a length-outlier factor for de-duplication.
- trimAl mode: standard (gappyout), gentle (automated1), or none.
- Outgroup families, one per line as "Family | mode | n" (mode = each_genus or top_species).
- Number of bootstrap replicates (default 1000).
- Optionally, "Restrict to project specimens": limit the ingroup to species selected from the Specimens page.

Restrict-to-specimens is the recommended mode: it guarantees every listed species that has a GenBank record for the marker is pursued, and writes each detected accession back onto the Specimens page.

Step A - Sequence Retrieval and Coverage Guarantee

The ingroup search runs the gene query in relaxed form (its "NOT (...)" exclusion is stripped) so that species whose only deposit is a combined ribosomal cassette (18S+ITS+28S in one record) are still recovered; per-marker trimming later keeps only the relevant region. When restricted to Specimens, the pipeline then guarantees coverage in three passes:

1. Recorded accessions: if a specimen already lists an accession for the marker, that exact record is fetched and used.
2. No-floor per-species rescue: any listed species not yet recovered is searched individually with no length filter, so even a short partial record is kept - added automatically.
3. Flexible search (manual): species with no record of the target marker trigger a broad organism-name search whose hits are queued on the Review Sequences screen for explicit accept/reject (never auto-added).

Only species for which GenBank holds no marker record at all remain missing, and those are reported by name. Every detected accession is written back to the Specimens page.

Step B - Quality, Orientation, Alignment, Trimming

- Each sequence is oriented by shared k-mer content against the longest record; reverse-complemented sequences are corrected and reported.

- Sequences with >5% ambiguous bases are flagged (kept, not removed).
- Alignment: MAFFT (--auto --adjustdirection) locally, or the equivalent Galaxy MAFFT tool.
- Trimming: trimAl in the selected mode, with a no-loss guarantee - if trimming would drop a sequence entirely, it falls back to a gentler mode or the untrimmed alignment, so trimming never removes a taxon.

Step C - Model Selection and NJ Preview

The best-fit substitution model is chosen with ModelTest-NG (by BIC), per partition in multi-fragment mode. A rapid neighbour-joining tree is produced as a PREVIEW ONLY, for a sanity check of rooting and obvious misplacements. Because it carries no bootstrap support, the NJ tree cannot be imported as the project tree.

Step D - Maximum-Likelihood Tree (RAxML-NG on Galaxy)

After you approve the sequence set, the alignment (and a partition file, if partitioned) is uploaded to usegalaxy.eu and RAxML-NG is run in all-in-one mode (--all): an adaptive ML search followed by non-parametric bootstrapping with autoMRE bootstopping and a fixed seed. Felsenstein bootstrap support (fbp) is requested explicitly and drawn on the best tree as node labels. Partitioned analyses estimate each fragment under its own model with proportionally linked branch lengths. The job is polled to completion, and the support-bearing tree is downloaded in preference to any support-free tree; a run returning no support values is flagged rather than accepted.

Step E - Review, Resubmit, Import, Rooting

- Revise & Resubmit: replace a sequence with another accession, reverse-complement, add, or remove a taxon, then rerun quality/alignment/trim/NJ before resubmitting.
- Import: the bootstrap-supported tree is imported into the matrix view; tip labels are parsed and any new species are added to the Specimens list.
- Rooting: specify one or more outgroup genera; the tree is rooted at their MRCA (ape via Rscript, or BioPython) with support values preserved, and saved as the project tree.

> *Only a bootstrap-supported tree can be imported, so every tree used downstream (matrix view, descriptions, exports) carries support values.*

15. Troubleshooting

"Database is locked" error

This occurs when multiple processes access SQLite simultaneously. The app has a 30-second timeout. Ensure only one instance of run.py is running. Delete any stale data/db.sqlite-journal file if the error persists.

Characters show all "?"

Click "Compute All Characters" on the project page. Characters require both landmarks and part boundaries to be confirmed before computation can proceed.

Species names not matching on import

Use the "Scan" button to preview how filenames are parsed before importing. The parser handles underscores, concatenated names, accession prefixes, and pipe-separated formats.

Thresholds seem wrong for my data

After importing new data, check the distribution of raw values via the Character Workshop. Adjust thresholds based on the actual data range for your taxon. The default thresholds are calibrated for Procrustes-normalized Gyrodactylidae data.

Gallery shapes have no colors

Part colors appear when a color legend is available. Ensure boundaries are confirmed for the structure type being viewed.

ImageJ macro cannot find images

Verify the input directory path in the macro setup dialog. The directory must contain image files (TIF, TIFF, JPG, JPEG, PNG, BMP, or GIF).

Phylogeny job fails during fetch with "HTTP Error 400"

NCBI Entrez rejects requests carrying an invalid API key with a 400 Bad Request. Verify the NCBI_API_KEY server environment variable holds a current, valid key (or remove it to run unauthenticated at a lower rate).

Specimens missing from the final alignment

Species reported as missing genuinely have no GenBank record for the chosen marker under that name. Check for placeholder ("sp.", "gen.") or misspelled names on the Specimens page; correct the name and rerun. Species that do have a record are recovered automatically by the no-floor per-species rescue.

Tree returned without bootstrap support

The pipeline requests Felsenstein bootstrap (fbp) support explicitly and prefers the support-bearing tree. If a run is flagged as returning no support, check the Galaxy history for the RAxML-NG job for a tool error before treating the tree as final.